

BIOS 755: Foundational concepts

Alexander McLain

January 15, 2026

Outline

Objectives of Longitudinal Analysis

Features of Longitudinal Analysis

Notation

Dependence and Correlation

Example

Sources of Variation

Introduction

Lecture goals:

- ▶ Give an overview of the objectives of longitudinal analysis and discuss features in the data.
- ▶ Longitudinal analysis estimates how individuals change throughout the study.
- ▶ Examine the factors related to individual differences over time.
- ▶ Review features of longitudinal study designs.
- ▶ Introduce notation.

Treatment of Lead-Exposed Children (TLC) Trial

Recall the TLC study (a balanced design)

- ▶ Exposure to lead during infancy is associated with substantial deficits in tests of cognitive ability
- ▶ Chelation treatment of children with high lead levels usually requires injections and hospitalization
- ▶ A new agent, Succimer, can be given orally
- ▶ Randomized trial examining changes in blood lead level during the course of treatment
- ▶ 100 children randomized to placebo or succimer
- ▶ Measures of blood lead level at baseline, 1, 4, and 6 weeks

Objectives of Longitudinal Analysis

- ▶ Defining feature of longitudinal study: two or more observations taken on (at least) some subjects.
- ▶ Multiple measurements over time allow assessment of within-individual changes in the response variable.
- ▶ Thus, some main goals are to:
 - ▶ characterize (a loaded word) the change in the response over time.
 - ▶ determine whether changes are related to exposures of interest

Longitudinal Analysis

- ▶ One way to achieve this is to look at “*change scores*” or “*difference scores*,” between pre- and post-treatment.
- ▶ There are different ways to form such a question
 1. are the *change scores* related to group.
 2. is the average difference in the change scores related to covariates.
 3. is the final score related to covariates (path analysis).
- ▶ 1 could be answered using a paired t-test.
- ▶ 2 could be answered using standard linear regression techniques (possibly adjusting for the pre-score and differences in time).
- ▶ 3 also could use linear regression (adjusting for the pre-score and differences in time).

Longitudinal Analysis

- ▶ The usefulness of change score analysis is limited to situations with two measurements; you will have multiple change scores with more than two measurements.
- ▶ Analyses with change scores should be used with caution as they
 - ▶ have been shown to lead to bias in observational (non-randomized) studies ([reference](#)), and
 - ▶ require complete data or multiple imputation.

Number of measurements

- ▶ The number of measurements can vary greatly from study to study and can be equally or unequally separated in time.
 - ▶ The amount of vigorous physical activity per hour for 14 days.
 - ▶ BMI measured in spring, fall, and summer for two years.
 - ▶ Height measured every 3 months until 3 years old, then every year thereafter.
- ▶ All of the above are considered to be **balanced**.

Number of measurements

- ▶ **Unbalanced** data are very common in the health sciences.
 - ▶ individuals will miss scheduled visits,
 - ▶ visits are not made exactly at the scheduled date,
 - ▶ timings are relative to a benchmark event (e.g., relative to seroconversion) or
 - ▶ visits are themselves random (common in retrospective data).
- ▶ Missing data are the rule, not the exception, so unbalanced data are the rule.

Data Notation

- ▶ Consider the following

Y_{ij} = the j th measurement taken on unit i .

where $i = 1, 2, \dots, N$ and $j = 1, 2, \dots, n$

- ▶ In the TLC data, each child had four measurements at baseline, 1, 4, and 6 weeks.
- ▶ Y_{ij} represents the **random** response of the i th child at measurement j , for $j = 1, 2, 3, 4$.
- ▶ t_{ij} is the time of the observation of the i th child at measurement j , for all i

$$t_{i1} = 0, \quad t_{i2} = 1, \quad t_{i3} = 4, \quad t_{i4} = 6$$

Random Vectors

- ▶ It is convenient to represent all observations for a specific unit as a **random vector**.
- ▶ For the TLC data each child has,

$$\mathbf{Y}_i = \begin{pmatrix} Y_{i1} \\ Y_{i2} \\ Y_{i3} \\ Y_{i4} \end{pmatrix}$$

the random vector for child i .

- ▶ Also use $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, Y_{i3}, Y_{i4})'$.
- ▶ In general, we have $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, \dots, Y_{in})'$.

Expectations and mean

- ▶ The mean, average, or “expectation” of each response is

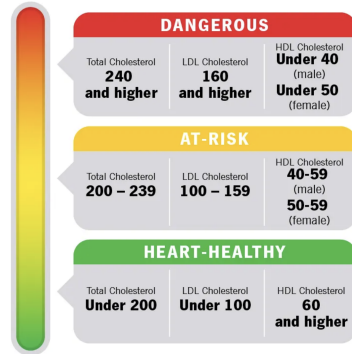
$$E(Y_{ij}) = \mu_{ij}$$

μ_{ij} is the *conditional* mean at the j th occasion (i.e., conditional on covariate values).

Dependence and Correlation Introduction

- ▶ Two variables are said to be independent if the behavior of one variable does not depend on the value of another variable.
- ▶ For example, LDL cholesterol, and sex are independent if the distribution of LDL cholesterol is the same for males and females.
- ▶ Longitudinal data methods do not assume that the observations are independent.
- ▶ For example, if I tell you my LDL cholesterol at baseline was 80, what is a plausible range for this value at 4 weeks?
- ▶ What if it was 165 at baseline?

Cholesterol Levels



Variance and Covariance

- ▶ The mathematical definition of variance

$$\text{var}(Y_{ij}) = E(Y_{ij} - \mu_{ij})^2 = \sigma_j^2$$

- ▶ The standard deviation is $\sqrt{\sigma_j^2} = \sigma_j$.
- ▶ **Covariance:** a measure of how two random variables vary together.
- ▶ Mathematically, this is expressed as,

$$\text{cov}(Y_{ij}, Y_{ik}) = E\{(Y_{ij} - \mu_{ij})(Y_{ik} - \mu_{ik})\} = \sigma_{jk}$$

Covariance Matix

- The covariance matrix of $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, \dots, Y_{in})'$

$$\text{Cov}(\mathbf{Y}_i) = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \dots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \dots & \sigma_n^2 \end{pmatrix} = \Sigma$$

- You will sometimes see $\sigma_j^2 = \text{var}(Y_{ij}) = \text{cov}(Y_{ij}, Y_{ij}) = \sigma_{jj}$.

Covariance to correlation

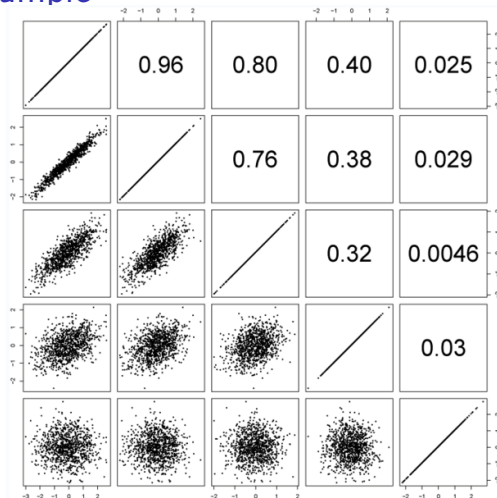
- ▶ The covariance values are hard to interpret since their magnitude depends on the variables' variance.
- ▶ Covariance is commonly standardized to correlation.
- ▶ The population correlation of two elements is

$$\rho_{jk} = \frac{\sigma_{jk}}{\sqrt{\sigma_j^2 \sigma_k^2}}$$

- ▶ Which gives the correlation matrix

$$\text{Corr}(\mathbf{Y}_i) = \begin{pmatrix} 1 & \rho_{12} & \dots & \rho_{1n} \\ \rho_{21} & 1 & \dots & \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} & \rho_{n2} & \dots & 1 \end{pmatrix}$$

Correlation Matix Example



Correlation in TLC

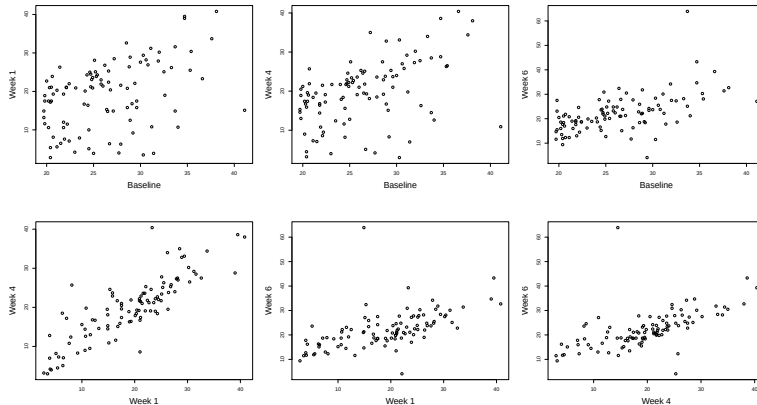


Figure: Correlation for 50 children in the placebo group

Estimated covariance & correlation matrices

- ▶ The estimated covariance matrix for the TLC

$$\text{Cov}(\mathbf{Y}_i) = \begin{pmatrix} 25.2 & 22.8 & 24.3 & 21.4 \\ 22.8 & 29.8 & 27.0 & 23.4 \\ 24.3 & 27.0 & 33.1 & 28.2 \\ 21.4 & 23.4 & 28.2 & 31.8 \end{pmatrix}$$

- ▶ The estimated correlation matrix for the TLC

$$\text{Corr}(\mathbf{Y}_i) = \begin{pmatrix} 1 & 0.83 & 0.84 & 0.76 \\ 0.83 & 1 & 0.86 & 0.76 \\ 0.84 & 0.86 & 1 & 0.87 \\ 0.76 & 0.76 & 0.87 & 1 \end{pmatrix}$$

Time plot for TLC

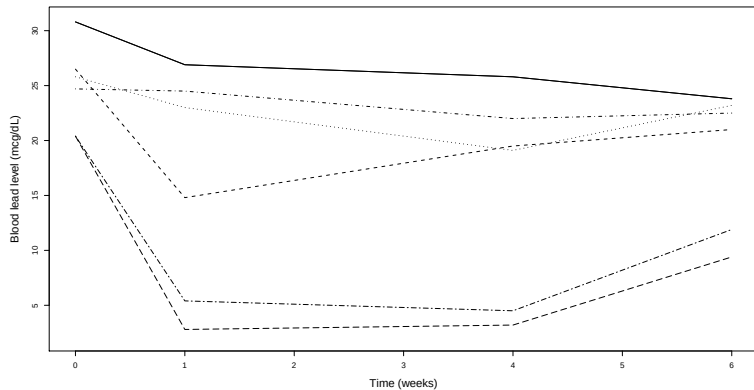


Figure: Time plot for six subjects

Correlation “truths”

1. the correlations are positive
2. the correlations often decrease with increasing time separation
3. the correlations between repeated measures rarely ever approach zero
4. the correlation between a pair of repeated measures taken very closely rarely approaches one.

Sources of variability

- ▶ Repeated measures let us watch change unfold, but different factors drive every data point.
- ▶ To model trajectories well, we need to know where the variability comes from—otherwise, we mis-attribute signals and inflate error bars.
- ▶ The three main factors that drive data are:
 - ▶ Differences Between People (between-individual heterogeneity)
 - ▶ Day-to-Day Fluctuations Within a Person (within-individual heterogeneity)
 - ▶ Imperfections in the Measurement Process (measurement error)

Differences Between People

- ▶ Each participant starts with a unique baseline (genes, lifestyle, environment).
- ▶ **Example:** Two adults can both lower their blood pressure with the same diet, yet begin 15 mm Hg apart.
- ▶ Some of the differences can be explained by adjusting for observed covariates (e.g., BMI), some cannot.
- ▶ Why it matters:
 - ▶ Explains wide vertical “spread” in spaghetti plots.
 - ▶ Induces correlation between the error terms
- ▶ Differences due to time-invariant exposures will result in a constant correlation between the error terms.

Day-to-Day Fluctuations Within a Person

- ▶ Even after accounting for baseline differences, individuals' trajectories aren't flat lines
- ▶ Bad sleep, stress, infections, and behavior swings, nudge values up and down.
- ▶ **Example:** A child's daily step count can oscillate $\pm 25\%$ around their weekly mean.
- ▶ Why it matters:
 - ▶ Creates "wiggle" in individual trajectories.
 - ▶ Encourages use of time-varying covariates or autoregressive terms.
- ▶ These can result in variance that changes over time.
- ▶ **Take-home:** Short-term biological rhythms and lifestyle shifts add their own layer of noise we must respect.

Imperfections in the Measurement Process

- ▶ Devices drift, lab assays vary, participants misreport.
- ▶ Example: Home glucose meters can differ by ± 10 mg/dL from venous draws.
- ▶ Why it matters:
 - ▶ Adds extra scatter that is not biologically meaningful.
 - ▶ Can hurt the power of analysis (i.e., our ability to see meaningful differences).
- ▶ Measurement error should be independent of time. So, it should not result in variance that changes over time.